Quantizing the Latent Communication Channel of a Recursive Multi-Agent System

A Preprint

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Abstract

Aggressively quantizing the latent communication channel of a recursive multi-agent language-model system—the dense activation vectors its agents exchange in place of text, here reduced to 2–4 bits per coordinate—preserves aggregate task accuracy but not the sequences of tokens the system generates, a question single-vector reconstruction error cannot settle. We insert the data-oblivious mean-squared-error core of TurboQuant into every latent link of the released RecursiveMAS system—without changing weights or prompts—and separately measure channel reconstruction, final-answer behavior, and token-trajectory fidelity across five task/tier cells on a single consumer GPU. Across the four non-confounded cells, paired greedy accuracy deltas stay within ± 2.4 pp (confidence intervals spanning zero), yet 4–10% of individual answers flip and 51–93% of greedy trajectories diverge within 128 tokens. This trajectory drift is strongly tier- and task-dependent—on code the larger constellation is far more robust, a +40.2 pp (95% CI $[+34.9, +45.7]$) gap stable across five quantizer rotations but nearly absent on math—and it is distributed across the channel: quantizing only the intra-model or only the cross-model links reproduces nearly all of it. A teacher-forced, position-aligned analysis shows the drift is margin-tipping—the channel perturbs every next-token distribution slightly but flips only low-margin decisions—and that the larger tier is more robust mainly because it attenuates that perturbation (76% of the gap) rather than because it decides more confidently (24%). The evidence supports aggregate answer robustness under aggressive fake quantization, not bit-exact losslessness, trajectory preservation, or a deployed bandwidth saving.

1 Introduction

Multi-agent language-model systems increasingly communicate through internal activation vectors. Such latent messages avoid decoding and re-encoding text, but they can contain thousands of floating-point coordinates and may be sent many times during recursive reasoning. In a distributed deployment, the communication channel can therefore become a systems bottleneck.

Quantization is well studied for model weights and the attention KV cache. A latent inter-agent message is a different target: it is consumed once, transformed by a learned cross-model adapter, and repeatedly replaced as agents refine an answer. Small local errors may be harmless, may accumulate, or may push greedy (deterministic, argmax) decoding across token-decision boundaries—changing which token is emitted next. These possibilities cannot be resolved from single-vector reconstruction error alone.

We study RecursiveMAS [Yang et al., 2026], a released recursive latent multi-agent system, and the data-oblivious MSE core of TurboQuant [Zandieh et al., 2026]. The intervention randomly rotates and scalar-quantizes every inter-agent vector. We measure three distinct outcomes: channel reconstruction (how faithfully the quantized vector is recovered), final-answer behavior (whether the system still solves the same problems), and token-trajectory fidelity (whether it emits the same sequence of tokens, step by step). Separating these matters: a system can preserve aggregate benchmark accuracy while changing which individual problems it solves and which reasoning text it generates.

Our central finding is a dissociation: at 2–4 bits the latent channel is answer-robust in aggregate yet does not preserve which problems are solved or which reasoning text is produced. Concretely, we contribute:

- the dissociation itself, quantified across five task/tier cells—paired greedy accuracy deltas within $\pm$2.4 pp (confidence intervals spanning zero) alongside 4–10% answer churn and 51–93% trajectory divergence (Figure 1);
- a same-task tier contrast—on code the scaled constellation is far more trajectory-robust than the light one (a +40.2 pp divergence gap)—shown to be task-specific (it nearly vanishes on math) and robust across five quantizer rotations, hence an association rather than a capacity law; with a link-type ablation showing the drift is not localized—quantizing only the inner (intra-model) or only the outer (cross-model) links each reproduces nearly the full divergence on both tiers;
- a teacher-forced, position-aligned mechanism for the drift: it is margin-tipping (flips concentrate at low-margin token decisions), and the larger tier's robustness is mostly attenuation of the perturbation (76% of the gap) rather than larger decision margins (24%);
- an instrumentation-only, reproducible pipeline—quantizing all RecursiveMAS latent links without changing the released models, prompts, or evaluation—with a corrected top-K trajectory analysis (primary-call pairing, length-censored first-divergence hazard) and documentation of two measurement failures: pre-Ampere bf16 numerical collapse and a greedy multiple-choice option bias.

This is an empirical study, not a new compression algorithm. We evaluate the MSE core of TurboQuant and omit its QJL inner-product residual.

2 Related work

Quantizing large language models. Most LLM quantization targets model weights—GPTQ [Frantar et al., 2023], AWQ [Lin et al., 2024], LLM.int8() [Dettmers et al., 2022], and SmoothQuant [Xiao et al., 2023]—or the attention key–value cache, where KIVI [Liu et al., 2024] pushes the cache to two bits. These objects are read repeatedly inside a single forward pass. The inter-agent latent message we compress is a different object: it is produced once, transformed by a learned cross-model adapter, and replaced as agents refine an answer, so its single-vector reconstruction error need not predict its downstream effect. Our quantizer is the data-oblivious mean-squared-error core of TurboQuant [Zandieh et al., 2026].

Multi-agent systems and latent communication. Multi-agent language-model systems improve reasoning by having models propose and critique answers, usually through text [Du et al., 2024]. A growing line replaces text with latent vectors: continuous-thought reasoning [Hao et al., 2024] and, between agents, systems that exchange hidden states directly—Interlat [Du et al., 2025], LatentMAS [Zou et al., 2025], and heterogeneous latent communication [Liu et al., 2026]—including the recursive system we study, RecursiveMAS [Yang et al., 2026]. These works design or learn the latent channel, and Interlat additionally learns to compress it into fewer reasoning steps for latency. We instead take a released recursive latent channel as given and ask a different question: how faithful its behavior remains under a training-free, data-oblivious, low-bit quantization of the transmitted vectors. Such latent channels are precisely what a low-bit transport would compress, which motivates the study.

Behavioral fidelity beyond accuracy. Compression is usually judged by aggregate benchmark accuracy. We instead separate final-answer behavior from token-trajectory fidelity, and report paired answer churn and greedy divergence; at the answer level we use a two-one-sided equivalence test [Lakens, 2017] rather than reading non-significance as equivalence. Our teacher-forced, position-aligned mechanism analysis builds on mechanistic-interpretability tooling [Nanda and Bloom, 2022]. We evaluate on Math500 [Hendrycks et al., 2021], MBPP+ [Austin et al., 2021], and MedQA [Jin et al., 2021].

3 System and intervention

RecursiveMAS. RecursiveMAS connects heterogeneous language-model agents through learned adapters and iterates latent messages across planning, solving, and critique stages. We pin the upstream source at commit `f95d512`. The completed matrix uses `sequential_light` and two `sequential_scaled` contrasts (MBPP+ and Math500). The upstream checkout is treated as read-only: the runner verifies its commit, creates a disposable source copy in the run directory, and instruments only that copy.

Quantizer. For $x \in \mathbb{R}^d$, TurboQuant-MSE normalizes x, applies a fixed Haar-random orthogonal transform, quantizes each transformed coordinate with a 2^b-level Lloyd–Max codebook for the concentrated marginal, applies the inverse transform, and restores the norm. We report

$$E_{\mathrm{rel}} = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2}$$

and cosine similarity at every wrapped adapter call. Quantization arithmetic is performed in fp32 and returned in the pipeline dtype.

Scope of compression. The ratios $32/b$ are nominal packed-payload ratios. Our experiments use fake quantization: the receiver consumes the same reconstructed vector that a low-bit transport would deliver, but we do not serialize codes or measure network latency. Metadata, rotation cost, and codec overhead are therefore outside the stated ratio.

4 Experimental design

4.1 Conditions

The primary study uses one RTX 5070 Ti 16 GB under WSL2, native bf16, seed 42, three recursive rounds, and $n = 250$ in every cell. The matrix contains:

- `sequential_light` on Math500, MBPP+, and MedQA;
- `sequential_scaled` on MBPP+ and Math500.

Earlier independent runs used Kaggle T4 fp32 for a sampled Math500 ladder and Modal A100 fp32 for controls and a depth sweep. They corroborate the broad Math500 result and expose dtype hazards, but are not the primary reproduction target.

Sampled ladders use the released sampling configuration at bits $\{0, 8, 4, 2\}$. Paired fidelity uses greedy decoding for REF ($b = 0$) and INT4 ($b = 4$), identical input order, and identical seed.

4.2 Outcome levels

Channel fidelity. We record cosine similarity, relative L2 error, norm ratio, and call counts.

Answer behavior. We report accuracy, paired delta, discordant counts, exact McNemar tests, bootstrap intervals, and churn: the fraction of samples whose correctness indicator changes. A two-one-sided equivalence test uses a pre-specified ± 2 percentage-point margin, but failure to establish equivalence is reported as inconclusive.

Trajectory behavior. Generation capture stores top-K logits and residual tail log-mass for up to 128 positions locally ($K = 256$). Only the fixed primary generate calls—exactly $\lceil n/\text{batch size} \rceil$ calls—are paired. Conditional answer-retry calls are excluded because REF and INT4 need not trigger them on the same examples. At each matched position, probabilities are reconstructed on the union of the two top-K supports. Mass assigned to newly exposed union tokens is subtracted from the residual tail before normalization, preventing double counting.

We report whether the greedy sequences diverge within the capture window, their common-prefix length, and KL/JS on the matched prefix. This KL remains approximate and selection-conditioned: once tokens diverge, the next-token contexts differ.

5 Results

Figure 1 previews the central finding: across the non-confounded cells, aggregate accuracy barely moves, individual answers change somewhat more often, and greedy trajectories change far more often still. The following subsections quantify each of the three levels.

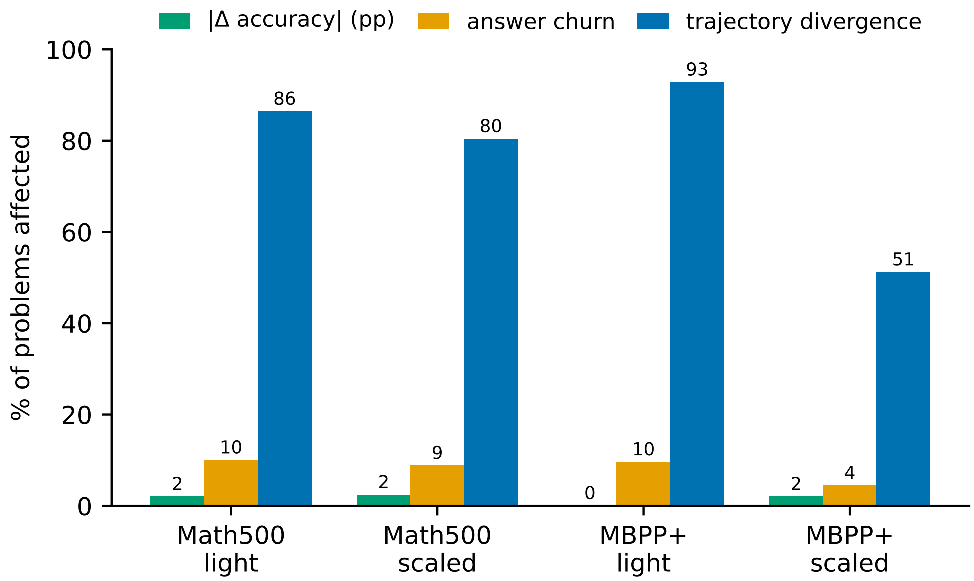

Figure 1: The central dissociation. For each non-confounded cell: the magnitude of the paired greedy accuracy change ($|\Delta\text{acc}|$, in percentage points), the fraction of problems whose correctness flips (answer churn), and the fraction of greedy trajectories that diverge from the unquantized run within 128 tokens—each as a percentage of problems. Aggregate accuracy is nearly unchanged ($\leq$ 2.4 pp) while 4–10% of answers churn and 51–93% of trajectories diverge: the 4-bit channel is answer-robust in aggregate but not trajectory-preserving. MedQA is omitted (its greedy reference is confounded; see Section 5.2).

Table 1: Relative squared reconstruction error of TurboQuant-MSE.

bits	synthetic ($d = 2048$)	real adapter ($d = 1536$)	reported rate
8	0.0001	0.0001	approximately 0
4	0.0095	0.0093	0.009
3	0.0345	0.0339	0.030
2	0.1175	0.1159	0.117

5.1 The real channel matches the expected distortion rate

Table 1 compares the real Solver adapter with synthetic Gaussian-on-sphere vectors. The 4-bit real-channel error, $E_{\text{rel}} = 0.0093$, matches the expected TurboQuant rate and has cosine about 0.995.

5.2 Five-cell local answer behavior

Table 2 extends the accuracy observation across both tasks and both tier contrasts. No sampled ladder is monotonically ordered by rate. In the four non-confounded cells (Math500/light, Math500/scaled, MBPP+/light, MBPP+/scaled), the paired greedy deltas are small and their 95% intervals span zero. Nevertheless, 4.4–10.0% of correctness labels change. Thus a near-zero net delta can conceal offsetting losses and gains.

The MedQA greedy row is confounded. REF selects option A about 46% of the time, whereas the sampled REF has no comparable bias. INT4 dither disrupts the bias and appears to improve accuracy. Because the reference condition is pathological, this row cannot estimate channel fidelity; the non-monotonic sampled ladder is the valid task-level observation.

Table 2: Local $n = 250$ answer results. Ladder order is REF/8/4/2 bits. Churn is paired greedy correctness churn.

cell	sampled ladder (%)	greedy REF/INT4 (%)	delta, 95% CI	churn
Math500 / light	77.6/76.0/78.8/78.0	76.8/78.8	+2.0 $[-2.0, +6.0]$	10.0%
Math500 / scaled	82.8/85.6/84.0/84.8	87.6/85.2	-2.4 $[-6.0, +1.2]$	8.8%
MBPP+ / light	32.8/33.6/38.8/34.8	36.4/36.4	0.0 $[-4.0, +4.0]$	9.6%
MBPP+ / scaled	70.8/73.6/72.0/71.2	74.4/72.4	-2.0 $[-4.8, +0.4]$	4.4%
MedQA / light	34.4/26.4/32.8/30.0	21.2/36.4	+15.2 $[+8.8, +21.6]$	30.4%

Table 3: Corrected local Tier-2 fidelity. Divergence is windowed at 128 captured positions; KL is approximate and restricted to matched prefixes.

cell	diverged	common prefix / 128	KL (nats)	channel cosine
Math500 / light	86.4%	53.7	0.079	0.9952
Math500 / scaled	80.4%	54.6	0.147	0.9953
MBPP+ / light	92.8%	35.8	0.113	0.9953
MBPP+ / scaled	51.2%	65.7	0.059	0.9953
MedQA / light	96.4%	32.4	0.045	0.9952

5.3 Corrected trajectory fidelity

Table 3 reports the corrected primary-call analysis. Individual channel cosine is almost identical in all cells, as expected from a data-oblivious quantizer at fixed rate. Downstream consequences are not identical. The light cells diverge in 86.4–96.4% of primary sequences within 128 positions. Scaled MBPP+ diverges in only 51.2%, retains a longer common prefix, and has lower matched-prefix KL—but scaled Math500 diverges in 80.4%, close to its light counterpart (86.4%), so the tier effect is large on MBPP+ and small on Math500.

The cleanest comparison holds task and local environment fixed: the light and scaled MBPP+ tiers, where scaling nearly halves divergence (Figure 2). Critically, this contrast does not reproduce on Math500, where the same light-to-scaled change moves divergence only 86.4% to 80.4% and even raises matched-prefix KL. A general capacity law would predict a comparable drop on both tasks; its absence on Math500 means the effect is task-specific. The result is compatible with larger or more capable models absorbing channel noise more effectively on some tasks, but capacity is not isolated: the tier also changes architecture, checkpoint family, baseline competence, margins, and generated length. Equal cosine likewise describes geometric reconstruction, not equal semantic perturbations. We therefore call this a task-conditioned tier association and treat a causal capacity claim as a hypothesis for replication.

Robustness to the quantizer rotation. To check that the MBPP+ contrast is not an artifact of one random rotation, we reran the INT4 capture for both tiers at five quantizer rotations (rotation seeds $42, 7, 17, 73, 101$), reusing the rotation-independent greedy REF. Per rotation, light divergence is 91.6–93.6% and scaled is 50.4–55.2% within 128 positions; the ranges never overlap. A problem-clustered bootstrap over the five rotations (problems resampled; rotations treated as repeated measures, not independent items) gives a light-minus-scaled contrast of +40.2 pp, 95% CI $[+34.9, +45.7]$ within 128 positions and +31.8 pp $[+25.9, +37.6]$ within the first 25 (the length-robust early window). The effect is thus robust to the quantizer rotation; only the generation seed and problem subset remain fixed.

Localizing the drift: inner versus outer links. Each tier exposes two link types—eight intra-model (inner) recursion adapters and eight cross-model (outer) adapters. To ask whether the trajectory drift is carried by a specific link type, we re-ran the MBPP+ INT4 capture quantizing exactly one type at a time, reusing the shared full-precision REF (Table 4). The drift is not localized: on both tiers, quantizing the inner links alone or the outer links alone each reproduces essentially the entire all-links divergence—every single-link value lies within 4 pp of its all-links value, and the cumulative first-divergence curves overlap. Either channel alone is therefore sufficient to reroute greedy decoding; the two are redundant rather than separable at this bit rate. The light/scaled robustness gap persists within each link type ($\sim$40 pp for both inner-only and outer-only), so the tier effect is a property of scale present in both channels, not something one link type carries. The one asymmetry is at the answer level: the small all-links accuracy cost on scaled (-2.0 pp)

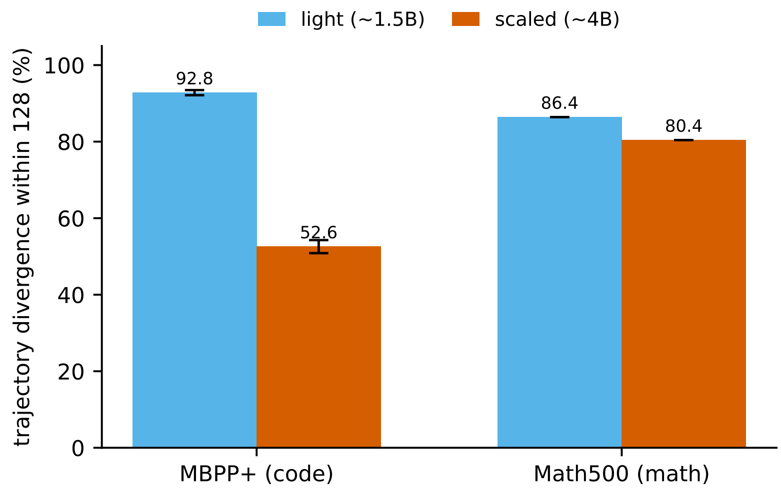

Figure 2: The tier contrast is task-specific. Greedy trajectory divergence within 128 tokens for the light (∼1.5B) and scaled (∼4B) tiers on MBPP+ (code) and Math500 (math). On code, scaling the constellation nearly halves divergence; on math the same change is small. Error bars on the MBPP+ bars are the standard deviation across five quantizer rotations (seeds $42, 7, 17, 73, 101$); Math500 is the single seed-42 rotation.

Table 4: The trajectory drift is not localized to a link type. MBPP+ INT4 greedy divergence within 128 tokens and final accuracy when quantizing all inter-agent links, only the eight inner (intra-model) links, or only the eight outer (cross-model) links. The all rows reproduce the headline numbers; each single-link condition recovers nearly the whole divergence on both tiers.

tier	quantized links	divergence within 128	accuracy	Δ acc. vs. REF
light	all	92.8%	36.4	±0.0
light	inner	88.8%	36.4	±0.0
light	outer	94.4%	35.6	−0.8
scaled	all	51.2%	72.4	−2.0
scaled	inner	50.4%	74.8	+0.4
scaled	outer	52.0%	72.8	−1.6

comes mainly from the outer links (outer-only −1.6 pp), whereas inner-only quantization leaves accuracy at the REF value—consistent with the channel-wide pattern of perturbed trajectories under preserved answers.

5.4 Mechanism: low-margin tipping under an attenuated perturbation

The free-running analysis counts that trajectories drift; to see why, we run an aligned teacher-forced capture: each decode position is forced along the full-precision REF tokens while the channel stays INT4-quantized, and we record the clean next-token logits, so REF and INT4 are compared position-by-position with no post-divergence confound. (Captures run at batch size 1, where forcing the full-precision REF reproduces the free-running REF exactly; the forced, window-truncated output is not a task metric.)

Three things hold on MBPP+ (Figure 3). First, the perturbation is universal but small: holding the prefix fixed, quantization shifts the next-token logits at essentially every position by a small amount (median top-1 logit shift 0.25 light, 0.125 scaled) and leaves the REF token ranked first 96–98% of the time. Second, arg-max flips concentrate at low margin: the flip probability is a strictly decreasing function of the REF top1–top2 logit margin (light 22.1% at margin <1 down to 0.2% above 16). The drift is a margin-tipping phenomenon—quantization changes the emitted token only where the decision is knife-edge. Third, decomposing the light−scaled flip-rate gap (+1.81 pp; 95% CI [+1.23, +2.40], problem-clustered bootstrap) shows the larger tier's robustness is not mainly a matter of deciding more confidently: the margin-distribution component is only +0.43 pp (24%), while +1.39 pp (76%) comes from the channel perturbing the larger model's logits less (the matched-margin flip rate is roughly halved). This is direct per-position evidence that the scaled

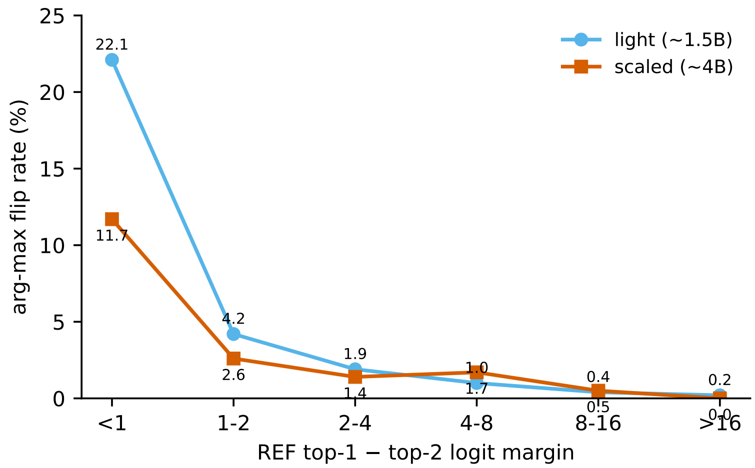

Figure 3: The drift is margin-tipping, and the larger tier absorbs the perturbation. Teacher-forced arg-max flip rate versus the REF top1–top2 logit margin on MBPP+. Flips concentrate at low margin on both tiers; the light curve sits above scaled at matched margin, and a decomposition attributes 76% of the tier gap to this attenuation and only 24% to scaled's larger-margin decisions.

Table 5: Independent historical Math500 sampled ladder (T4 fp32, $n = 250$).

bits	nominal compression	accuracy	delta versus REF
REF	1×	75.2%	—
8	4×	78.4%	+3.2 pp
4	8×	76.8%	+1.6 pp
2	16×	75.2%	0.0 pp

constellation absorbs the latent-channel noise, and it sharpens the "larger models absorb channel noise" reading: absorption dominates decision-confidence.

Task-specificity of the attenuation. Repeating the teacher-forced capture on Math500—where the free-running tier gap is near zero—separates the universal part of the mechanism from the task-specific part. Margin-tipping is universal: the flip-vs-margin gradient holds on Math500 for both tiers (light 25.7% at margin <1 down to 0.1%). The tier gap, however, does not generalize. The light−scaled flip-rate gap is -0.88 pp (95% CI $[-1.43, -0.34]$)—about half the magnitude of the MBPP+ gap and slightly reversed in sign—and the attenuation component collapses by nearly an order of magnitude, from $+1.39$ pp to -0.18 pp. The larger constellation's noise-absorption advantage is therefore task-conditioned, not a general capacity property, mirroring the ∼40 pp (MBPP+) versus ∼0 (Math500) free-running contrast.

5.5 Independent cloud Math500 corroboration

An earlier fp32 T4 ladder at $n = 250$ is flat within its uncertainty (Table 5, Figure 4). At 2 bits, REF and the quantized system both solve 188/250 problems. These unpaired stochastic runs support "no detected degradation," not exact equality, and provide an independent hardware/dtype corroboration of the local Math500 pattern.

6 Measurement failures and reproducibility lessons

Hardware precision. On pre-Ampere GPUs, the released bf16 configuration silently collapses RecursiveMAS recursive inference to roughly 30–35% Math500 accuracy. Forcing fp32 restores the baseline. A separate A100 run initially showed a 19-point quantized drop because repeated bf16–fp32 boundary casts were introduced by the instrumentation. A dtype-coherent pipeline removes the artifact. These are pipeline failures, not compression results.

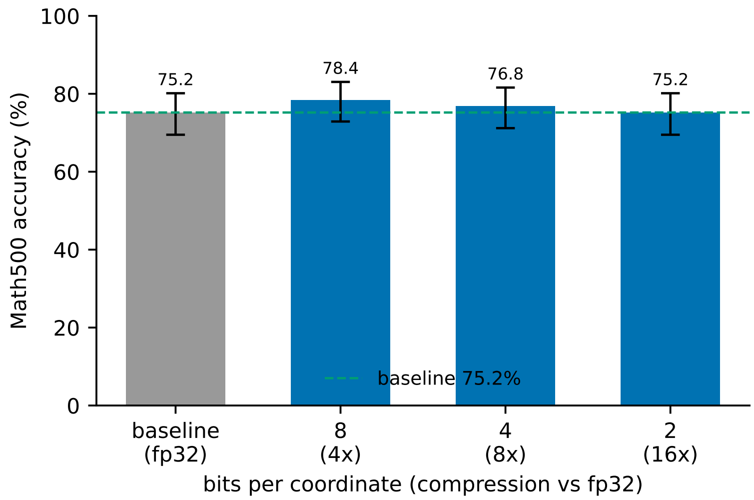

Figure 4: Independent historical corroboration: T4 fp32 Math500 sampled accuracy versus bit rate ($n = 250$, Wilson 95% intervals). The ladder is non-monotonic and every interval overlaps the baseline; these unpaired sampled runs support "no detected degradation," not equivalence.

Conditional generation calls. The first Tier-2 local analysis paired every captured `generate()` call. Answer extraction can trigger a retry conditionally, so after REF and INT4 diverge those calls are not structurally aligned. Restricting analysis to the fixed primary calls removes this bias.

Top-K residual mass. When a token appears in only one captured top-K, its approximated probability in the other distribution must be deducted from that distribution's residual tail. Failing to do so double counts mass and inflates KL. The corrected analyzer and synthetic regression tests implement this accounting. Historical cloud KL values produced by the legacy estimator are retained for auditability but are not compared numerically with Table 3.

Reproducibility. The instrumentation, resumable orchestrators, analyzers, and per-result expected values are released; the upstream RecursiveMAS is pinned at commit `f95d512` and treated as read-only. A SHA256-plus-provenance manifest of every raw capture (paths relative to the capture root, no secrets) together with the runtime environment is committed, so a regenerated set can be checked byte-for-byte. Step-by-step commands for every table and figure are in the repository's reproducibility guide.

7 Discussion

The central result is a dissociation. Aggressive channel quantization produces nearly constant aggregate accuracy at the resolution of these experiments, yet it changes individual answers and most light-tier greedy trajectories. "Answer preserving" is therefore meaningful only as a population-level shorthand; it must not be read as preserving every answer or every reasoning trace.

Symmetrically, the headline divergence rate should not be read as 51–93% of behavior changing. Conditioning the outcome on whether the trajectory diverged, 89–91% of diverged trajectories still produce the same final outcome (unit-test pass/fail on MBPP+, boxed answer on Math500), and the per-problem probability that divergence coincides with an outcome flip is only 8.6–10.6% across cells. The greedy token path changes wholesale, but it rarely crosses a task-relevant decision boundary—the operationally interesting object is exactly that small, low-margin subset where it does (Figure 3). The $\sim$90% preserved fraction is moreover stable across tasks and tiers despite the 51–93% spread in raw divergence, consistent with a margin-tipping mechanism in which the per-decision flip probability, not the cumulative path divergence, governs the outcome.

The MBPP+ result weakens a simple redundancy explanation. Code tokens are themselves the evaluated output, yet the light and scaled sampled ladders remain flat. The scaled tier's reduced churn and trajectory divergence on MBPP+ are now explained mechanistically by the teacher-forced analysis (Figure 3): the larger constellation flips fewer tokens both because it decides with larger margins and—dominantly (76% of the gap)—because it attenuates the channel perturbation reaching its output logits, rather than because of a selection effect created by matched prefixes. Because the same tier change leaves Math500 divergence almost unchanged, this noise-absorption must itself be task-dependent.

The nominal compression ratios are promising but incomplete. A real system must encode codebook indices, norms, seeds or transform identifiers, and packet metadata; it must also pay rotation and codec latency. The present work establishes behavioral fidelity of reconstructed messages, not end-to-end throughput.

8 Limitations and next steps

All local cells use one machine and a single generation seed. The {math, code} $\times$ {light, scaled} 2×2 is now crossed, the MBPP+ tier contrast is additionally confirmed across five quantizer rotations and decomposed into inner- and outer-link contributions; but the generation seed and problem subset remain singular. MedQA greedy is confounded, and MBPP+ contains only 378 available problems. Confidence intervals still allow effects of several percentage points. The trajectory window is 128 positions and local capture uses $K = 256$; the teacher-forced mechanism analysis covers MBPP+ and a Math500 control at one generation seed (the matched-prefix KL it complements remains a selection-conditioned estimate). We evaluate fake quantization and the TurboQuant MSE core only.

The highest-value next steps are: (i) sweep the generation seed and problem subset on MBPP+ (the five-quantizer-rotation matrix is complete and confirms the light/scaled gap, +40.2 pp [+34.9, +45.7]); (ii) extend the aligned mechanism analysis to a second task and to additional bit rates (2/3/4/6/8), testing whether the margin-tipping/attenuation account holds off MBPP+ and whether the two channels separate at higher rates; (iii) implement the QJL residual; (iv) test a tool-sensitive deliberation topology and a high-baseline non-math task; and (v) measure serialized bytes, latency, wall-clock time, and VRAM. Confirmatory claims should pre-specify the primary estimand, equivalence margin, seed pooling, and multiple-comparison policy.

9 Conclusion

The latent channel of RecursiveMAS tolerates aggressive data-oblivious fake quantization well in aggregate accuracy across the completed cells, yet that robustness coexists with real behavioral change: individual answers churn and greedy trajectories often diverge. A position-aligned analysis shows the drift is margin-tipping—the channel perturbs every token decision slightly but flips only the low-margin ones—and that the scaled MBPP+ tier is more trajectory-robust mainly because it attenuates the perturbation, and only secondarily because it decides with larger margins. This tier contrast is robust across five quantizer rotations but task-specific (it does not reproduce on Math500), so the noise-absorption it reflects is itself task-conditioned, not a capacity law. The practical conclusion is precise: low-bit latent communication is a credible systems direction, but claims of losslessness require stronger statistics, broader tasks and architectures, and real transport measurements.

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